# DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

**Course: B. Tech.** **Branch: Electrical Engineering** **Semester: VII**

**Subject Code & Name: Electrical Drives (BTEEC703)**

**Max Marks: 60** **Date: 22/08/2022** **Duration: 3.45 Hrs.**

***Instructions to the Students:***

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

| | | (Level/CO) | Mark |
|---|---|---|---|
| **Q.1** | **Solve Any Two of the following.** | | |
| **A)** | **Explain in detail the various types of electric drives** | Understanding | **06** |
| **B)** | **Draw block diagram of electrical drives and explain its different parts.** | Understanding | **06** |
| **C)** | **Explain four quadrant operation of motor driving hoist load.** | Applying | **06** |
| **Q.2** | **Solve Any Two of the following.** | | |
| **A)** | **Explain the choice of selection of the motor for different loads.** | Understanding | **06** |
| **B)** | **Write a brief note on classes of duty for on electric motor in on electric drive** | Understanding | **06** |
| **C)** | **Explain closed loop speed control of multi motor drives.** | Understanding | **06** |
| **Q.3** | **Solve Any Two of the following.** | | |
| **A)** | **A 200 V, 875 rpm, 150 A separately excited de motor has an armature resistance of 0.06 Ω. It is fed from a single phase fully-controlled rectifier with an ac source voltage of 220 V, 50 Hz Assuming continuous conduction, calculate**<br>**(1) firing angle for rated motor torque and 750 rpm**<br>**(2) firing angle for rated motor torque and (-500) rpm.** | Understanding | **06** |
| **B)** | **Explain the single-phase full wave-controlled rectifier control of DC separately excited drive with waveforms.** | Applying | **06** |

| | | | | |
|---|---|---|---|---|
| **C)** | **Explain breaking technique of DC motor in detail :**<br>**1) dynamic breaking 2) Regenerative breaking** | Applying | **06** |

**Q.4 Solve Any Two of the following.**

| | | | |
|---|---|---|---|
| **A)** | **Explain slip power recovery scheme for induction motor drive**<br>**1) Static scherbius drive**<br>**2) Static Kramer drive** | Understanding | **06** |
| **B)** | **Explain operation of VSI fed induction motor drive.** | Understanding | **06** |
| **C)** | **Discuss variable frequency control of induction motor drive. Draw relevant speed torque character sties.** | Understanding | **06** |

**Q.5 Solve Any Two of the following.**

| | | | |
|---|---|---|---|
| **A)** | **Draw the block diagram of variable frequency control of multiple synchronous motor drive and explain in brief.** | Understanding | **06** |
| **B)** | **Explain in detail drives for cement mills application.** | Understanding | **06** |
| **C)** | **Explain in brief drives required for various stages in textile mills** | Applying | **06** |

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